Assignment 10: Exploratory Data Analysis (EDA)

Instructions

- The data file **healthcare_dataset_missing.csv** has been uploaded. Perform the following EDA and modelling tasks on this data using Python.
- You are required to submit a single ZIP file named "**yourname_Assignment_10.zip**" This ZIP file must contain the following files:-
    - **yourname_eda_Assignment_10.ipynb** : This file should contain the code and the respective plots. Answer the questions in order. For each question add a markdown cell prior to the code indicating the question number.
    - **yourname_analysis_Assignment_10.pdf** : This file should contain the conclusions/analysis for the questions that expect this. Also add the plots obtained in this file.

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1. Carry out the following data understanding and integrity checks
    a. How many rows and columns are present in the data?
    b. How many rows and how many columns have missing data?
2. Identify the level of measurement associated with each column
3. Create appropriate Descriptive Statistics information for every column, based on it's level of measurement, and organize all this data in Tables for easy analysis and decision making.
    a. Analyze the descriptive statistics thus generated and make your initial conclusions
4. Is this data a time-series data? Do you expect any trends in each of the columns? Can you make any conclusions by creating a scatter plot of values of each column?
5. Based on the information available so far, what will be your strategy for dealing with the missing values? Make some initial decisions at this stage.
6. Analyze the Age column with the help of appropriate visualizations. What can you conclude? Are there any missing values in this column? How will you handle them? Why?
7. Create histograms and KDE plots (find out what KDE plots represent) for all the columns and analyze them. Any significant conclusions?
8. Create box plots for all the columns, individually, and analyze them. What are your conclusions?
9. Create box plots for all the numerical columns on a common scale, and in a single plot, and analyze this plot. What do you observe? What are its implications?
10. To understand the relationships between the columns, create the following:
    a. Pairwise scatter plots
    b. Heatmap of pairwise correlation coefficients
    c. What are your conclusions based on these plots?
11. Based on all the analysis so far, what is your final conclusion regarding the handling of missing values? Implement your decision!

12. Using a suitable pipeline approach (using sklearn pipeline), implement scaling, transformation and Linear Regression together to predict the Billing Amount.

    You would follow below pointers for the preprocessing

    - Drop irrelevant columns (such as Name, Doctor, Hospital),
    - Convert the Date of Admission and Discharge Date into datetime format,
    - Create a new feature called Length of Stay as the difference (in days) between Discharge Date and Date of Admission. Remove the original date columns after feature creation.
    - Encode all categorical variables appropriately.
    - Split the dataset into training and testing sets,

    In the end report the performance metrics for both train and test data.

13. Scale the numerical columns in the dataset using an appropriate technique (such as Min-Max Scaling or Standardization). After scaling, re-create a common-scale box plot for all numerical variables in a single plot.

14. Compare this plot with the earlier box plot (before scaling) and analyze how scaling has affected the distribution, spread, and presence of outliers. Interpret your observations. Additionally, create correlation heatmaps before and after scaling, and compare them. What are your observations?